1. Appendix 1.

EES Code for Modeling a Conventional MSSG with Equation Derivation
This annex contains the following elements:

1. The original EES code, preserving all comments exactly as they were written.
2. The full derivation of the key equations used in the heat exchangers (superheater, reheater, boiler, and preheater), preceded by a clear definition of each variable.
3. Finally, the numerical results obtained from the calculations are presented, organized into tables or an executive summary.

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"!EVALUATION OF THE CONVENTIONAL MSSG (SUPERHEATER-REHEATER-BOILER-PREHEATER)"

"Define the bookkeeping method"
turb_in = 1      "turbine inlet"
ph_in = 36       "Preheater HX inlet"
boil_in = 49     "Boiler inlet"
sh_in = 50       "Superheater HX inlet"
RHC = 4          "Main steam inlet to valve box"
RHH = 5          "Reheater inlet"

"!Original heat transfer fluid data"
T_htf_hot_sh = 565.55 + 273.15 [K]
T_htf_cold = 288 + 273.17 [K]
T_htf_hot_rh = 565.55 + 273.15 [K]

"!Original conditions considering a pressure drop in the boiler"
T[turb_in] = 811.15 [K]
P[turb_in] = 16700000 [Pa]
h[turb_in] = Enthalpy(Steam_IAPWS, T = T[turb_in], P = P[turb_in])

P[sh_in] = 18214192 [Pa]
T[sh_in] = T_boil
h[sh_in] = Enthalpy(Steam_IAPWS, T = T[sh_in], P = P[sh_in])

P[boil_in] = 18214192 [Pa]
T_boil = T_sat(Steam_IAPWS, P = P[boil_in])
T[boil_in] = T_boil - 0.01 [K]
h[boil_in] = Enthalpy(Steam_IAPWS, T = T[boil_in], P = P[boil_in])

T[ph_in] = 524.6
P[ph_in] = 18481616 [Pa]
h[ph_in] = Enthalpy(Steam_IAPWS, T = T[ph_in], P = P[ph_in])

m_dot_steam = 84.84 [kg/s]

"!Reheater inlet"
P[RHC] = 4198761 [Pa]
T[RHC] = 615.4 [K]
h[RHC] = Enthalpy(Steam_IAPWS, T = T[RHC], P = P[RHC])

"!Reheater outlet"
P[RHH] = 3779027 [Pa]
T[RHH] = 811.2 [K]
h[RHH] = Enthalpy(Steam_IAPWS, T = T[RHH], P = P[RHH])

m_dot_RH = 76.17 [kg/s]

"!Heat transfer fluid"
HTF$ = 'Salt(60NaNO3_40KNO3)'
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"!CALCULATION OF DESIGN UA OF HEAT EXCHANGING EQUIPMENT (REF)"

"!HTF mass flow in the Superheater"
T_htf_med_sh = (T_htf_hot_sh + T_htf_cold) / 2
cp_htf_avg1 = SpecHeat(HTF$, T = T_htf_med_sh) "Average HTF specific heat between SH inlet and outlet"
m_dot_htf1 = ((h[turb_in] - h[ph_in]) * m_dot_steam) / (cp_htf_avg1 * (T_htf_hot_sh - T_htf_cold))

"!Heat absorbed by steam in the superheater"
q_sh = h[turb_in] - h[sh_in] "Specific heat flux absorbed by steam in the SH"
Q_dot_sh = q_sh * m_dot_steam "Heat flux absorbed by steam in the SH"
cp_s_sh = SpecHeat(Steam_IAPWS, T = (T[turb_in] + T[sh_in]) / 2, P = P[sh_in]) "Specific heat of steam in the SH"
C_dot_s_sh = cp_s_sh * m_dot_steam "Heat capacity rate of steam in the SH"

T_htf_avg_sh = (T_htf_hot_sh + T_htf_out_sh) / 2
cp_htf_sh = SpecHeat(HTF$, T = T_htf_avg_sh) "Average HTF specific heat in the SH"
C_dot_hft_sh = cp_htf_sh * m_dot_htf1 "HTF heat capacity rate in the SH"

Q_dot_htf_sh = C_dot_hft_sh * (T_htf_hot_sh - T_htf_out_sh)

"!Superheater sizing"
C_dot_min_sh = MIN(C_dot_s_sh, C_dot_hft_sh) "Minimum heat capacity rate"
C_dot_max_sh = MAX(C_dot_s_sh, C_dot_hft_sh) "Maximum heat capacity rate"
C_R_sh = C_dot_min_sh / C_dot_max_sh "Heat capacity ratio"

Q_dot_max_sh = C_dot_min_sh * (T_htf_hot_sh - T[sh_in])

"!NTU–effectiveness relationship for counterflow using EES HX library: HX(TypeHX$, P, C1, C2, Return$)"
epsilon_sh = Q_dot_sh / (C_dot_min_sh * (T_htf_hot_sh - T[sh_in])) "Effectiveness"

"!Actual superheater outlet temperature"
T_htf_out_sh = T_htf_hot_sh - (Q_dot_sh / C_dot_hft_sh) "Actual HTF inlet to boiler, replace assumed value"

"!HTF mass flow in the Reheater"
T_htf_med_rh = (T_htf_hot_rh + T_htf_cold) / 2
cp_htf_avg2 = SpecHeat(HTF$, T = T_htf_med_rh) "Average HTF specific heat between RHC inlet and RHH outlet"
m_dot_htf2 = ((h[RHH] - h[RHC]) * m_dot_RH + (h[boil_in] - h[ph_in]) * m_dot_steam) / (cp_htf_avg2 * (T_htf_hot_rh
- T_htf_cold)) "HTF mass flow"

"!Heat absorbed by steam in the reheater"
q_RH = h[RHH] - h[RHC] "Specific heat flux absorbed by steam in the RH"
Q_dot_RH = q_RH * m_dot_RH "Heat flux absorbed by steam in the RH"
cp_s_RH = SpecHeat(Steam_IAPWS, T = (T[RHH] + T[RHC]) / 2, P = P[RHC]) "Specific heat of steam in the RH"
C_dot_s_RH = cp_s_RH * m_dot_RH "Heat capacity rate of steam in the RH"

T_htf_avg_rh = (T_htf_hot_rh + T_htf_dren_rh) / 2
cp_htf_RH = SpecHeat(HTF$, T = T_htf_avg_rh) "Average HTF specific heat in the RH"
C_dot_hft_RH = cp_htf_RH * m_dot_htf2 "HTF heat capacity rate in the RH"

Q_dot_htf_RH = C_dot_hft_RH * (T_htf_hot_rh - T_htf_dren_rh)

"!Reheater sizing"
C_dot_min_RH = MIN(C_dot_s_RH, C_dot_hft_RH) "Minimum heat capacity rate"
C_dot_max_RH = MAX(C_dot_s_RH, C_dot_hft_RH) "Maximum heat capacity rate"
C_R_RH = C_dot_min_RH / C_dot_max_RH "Heat capacity ratio"

Q_dot_max_RH = C_dot_min_RH * (T_htf_hot_rh - T[RHC])

"!NTU–effectiveness relationship for counterflow using EES HX library: HX(TypeHX$, P, C1, C2, Return$)"
epsilon_RH = Q_dot_RH / (C_dot_min_RH * (T_htf_hot_rh - T[RHC])) "Effectiveness"

"Alternative method"
ARH = epsilon_RH - 1
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BRH = epsilon_RH * C_R_RH - 1
CRH = ln(ARH / BRH)
DRH = C_R_RH - 1
NTU_RH = CRH / DRH

UA_RH = NTU_RH * C_dot_min_RH "Heat transfer area coefficient of the RH"

"!Actual reheater outlet temperature"
T_htf_dren_rh = T_htf_hot_rh - (Q_dot_RH / C_dot_hft_RH) "Actual HTF inlet to boiler, replace assumed value"

"!Boiler inlet temperature"
Cp_htf_out_sh = Cp('Salt(60NaNO3_40KNO3)', T = T_htf_out_sh)
Cp_htf_dren_rh = Cp('Salt(60NaNO3_40KNO3)', T = T_htf_dren_rh)
T_htf_boil_in = ((m_dot_htf1 * Cp_htf_out_sh * T_htf_out_sh) + (m_dot_htf2 * Cp_htf_dren_rh * T_htf_dren_rh)) /
((m_dot_htf1 + m_dot_htf2) * ((Cp_htf_out_sh + Cp_htf_dren_rh) / 2))

"!HTF mass flow in the boiler and preheater"
m_dot_htf = m_dot_htf1 + m_dot_htf2

"!Heat absorbed by water in the boiler"
q_boiler = h[sh_in] - h[boil_in] "Specific heat flux absorbed by water in the boiler"
Q_dot_boiler = q_boiler * m_dot_steam "Heat flux absorbed by water in the boiler"

"!Boiler sizing"
cp_htf_boiler = SpecHeat(HTF$, T = (T_htf_boil_in + T_htf_ph_in) / 2) "Average HTF specific heat in the boiler"
C_dot_hft_boiler = m_dot_htf * cp_htf_boiler "HTF heat capacity rate in the boiler"

Q_dot_htf_boiler = C_dot_hft_boiler * (T_htf_boil_in - T_htf_ph_in)

Q_dot_max_boiler = C_dot_hft_boiler * (T_htf_boil_in - T[boil_in])

epsilon_boiler = Q_dot_boiler / (C_dot_hft_boiler * (T_htf_boil_in - T[boil_in])) "Effectiveness"
NTU_boiler = -ln(1 - epsilon_boiler) "NTU for the boiler"
UA_boiler = NTU_boiler * C_dot_hft_boiler "Heat transfer area coefficient of the boiler"

"!Heat absorbed by water in preheater"
q_ph = h[boil_in] - h[ph_in] "Specific heat flux absorbed by water in the PH"
Q_dot_ph = q_ph * m_dot_steam "Heat flux absorbed by water in the PH"
cp_w_ph = SpecHeat(Steam_IAPWS, T = (T[boil_in] + T[ph_in]) / 2, P = P[boil_in]) "Specific heat of water in the PH"
C_dot_w_ph = cp_w_ph * m_dot_steam "Heat capacity rate of water in the PH"

T_htf_avg_ph = (T_htf_ph_in + T_htf_cold) / 2
cp_htf_ph = SpecHeat(HTF$, T = T_htf_avg_ph) "Average HTF specific heat in the PH"
C_dot_hft_ph = m_dot_htf * cp_htf_ph "HTF heat capacity rate in the PH"

Q_dot_htf_ph = C_dot_hft_ph * (T_htf_ph_in - T_htf_cold)

"!Preheater sizing"
C_dot_min_ph = MIN(C_dot_w_ph, C_dot_hft_ph) "Minimum heat capacity rate"
C_dot_max_ph = MAX(C_dot_w_ph, C_dot_hft_ph) "Maximum heat capacity rate"
C_R_ph = C_dot_min_ph / C_dot_max_ph "Heat capacity ratio"

Q_dot_max_ph = C_dot_min_ph * (T_htf_ph_in - T[ph_in])

"!Actual preheater inlet temperature"
"!Do not use; HTF inlet to PH is determined by boiler outlet"
T_htf_ph_in = T_htf_boil_in - (Q_dot_boiler / C_dot_hft_boiler) "Actual HTF inlet to PH, replace assumed value"

"!NTU–effectiveness relationship for counterflow using EES HX library: HX(TypeHX$, P, C1, C2, Return$)"
epsilon_ph = Q_dot_ph / (C_dot_min_ph * (T_htf_ph_in - T[ph_in])) "Effectiveness"

"Alternative method"
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Aph = epsilon_ph - 1
Bph = epsilon_ph * C_R_ph - 1
Cph = ln(Aph / Bph)
Dph = C_R_ph - 1
NTU_ph = Cph / Dph

UA_ph = NTU_ph * C_dot_min_ph "Heat transfer area coefficient of the PH"
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"!NUMERICAL RESULTS OBTAINED"

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Aph=-0.18
ARH=-0.1039
boil_in=49
Bph=-0.6268
BRH=-0.5169
Cph=-1.248
cp_htf_avg1=1.521 [kJ/kg·-·K]
cp_htf_avg2=1.521 [kJ/kg·-·K]
cp_htf_boiler=1.521 [kJ/kg·-·K]
Cp_htf_dren_rh=1.526
Cp_htf_out_sh=1.527
cp_htf_ph=1.506 [kJ/kg·-·K]
cp_htf_RH=1.536 [kJ/kg·-·K]
cp_htf_sh=1.536 [kJ/kg·-·K]
cp_s_RH=2.330 [kJ/kg·-·K]
cp_s_sh=3.745 [kJ/kg·-·K]
cp_w_ph=5.477 [kJ/kg·-·K]
CRH=-1.604
C_dot_hft_boiler=1031
C_dot_hft_ph=1021
C_dot_hft_RH=329.2
C_dot_hft_sh=711.5
C_dot_max_ph=1021
C_dot_max_RH=329.2
C_dot_max_sh=711.5
C_dot_min_ph=464.6
C_dot_min_RH=177.5
C_dot_min_sh=317.7
C_dot_s_RH=177.5
C_dot_s_sh=317.7
C_dot_w_ph=464.6
C_R_ph=0.4552
C_R_RH=0.5392
C_R_sh=0.4466
Dph=-0.5448
DRH=-0.4608
epsilon_boiler=0.6233 [-]
epsilon_ph=0.82 [-]
epsilon_RH=0.8961 [-]
epsilon_sh=1.155 [-]
HTF$='Salt(60NaNO3_40KNO3)'
m_dot_htf=677.6 [kJ/s]
m_dot_htf1=463.3 [kg/s]
m_dot_htf2=214.4 [kg/s]
m_dot_RH=76.17 [kg/s]
m_dot_steam=84.84 [kg/s]
NTU_boiler=0.9764
NTU_ph=2.29
NTU_RH=3.481
ph_in=36
q_boiler=759.6
Q_dot_boiler=64448 [kJ/s]
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Q_dot_htf_boiler=64448 [kJ/s]